

The DB SQL GUI

The DB SQL GUI (DB_SQL_main.py) lets you query a relational database with SQL: it builds an SQLite database from a set of data files, then lets you write, run, save and re-import queries, and export the results.

This TIPS explains what input the GUI needs, where the database it builds is saved, and what to do when the GUI opens with everything greyed out.

1. What input the GUI needs

This is the single most common source of confusion. The DB SQL GUI needs SQL/SQLite data — not text files, and not PC-ACE data as such. It accepts EITHER of the following:

- **An input directory of data files:** a folder holding a set of .xlsx (preferred) or .csv data files that share overlapping ID fields. The GUI builds an SQLite database from them automatically.
- **An existing SQLite database:** chosen directly, if you already have one. No input directory of data files is then required.

A directory of plain text documents (a normal NLP Suite corpus, e.g. a folder of .txt files) is NOT valid input. There is nothing for the GUI to build a database from, so every widget stays disabled.

If the GUI opens with everything greyed out

That means no SQLite database is available yet. The GUI will tell you so, and the reason is almost always one of:

- The selected input directory contains no .xlsx or .csv files (for example, a text corpus folder was selected).
- No input directory or no output directory has been set.

Fix it by selecting an input directory that contains data files, or by selecting an existing SQLite database. The RUN button and the query widgets enable themselves as soon as a database and an output directory are both set.

2. Where the database is built and saved

When you change the input directory, the GUI builds the SQLite database automatically — you do not need to press anything. The database is saved INSIDE the input directory (not the output directory), named after that folder:

```
<input directory>\<input directory name>.sqlite
```

For example, an input directory named PCACE-lynching_xlsx produces:

```
PCACE-lynching_xlsx\PCACE-lynching_xlsx.sqlite
```

Building can take several minutes on a large database; a notice appears while it runs, and the window title shows "Loading ..." until it finishes.

Reuse and automatic rebuilding

The GUI does not rebuild the database every time. It records a version stamp in a small file, _sqlite_version.txt, next to the database:

- If a database already exists for that folder AND its version matches, it is reused as-is (fast). You will see a brief "Reusing existing SQLite database" notice.
- If the version is out of date (because the column mappings changed in a newer NLP Suite release), the database is rebuilt automatically and the notice says so.

To force a rebuild from scratch, delete the .sqlite file (or _sqlite_version.txt) from the input directory and re-select the folder.

The "SQL queries" subdirectory

The GUI creates a subdirectory named "SQL queries" inside your input directory. Queries you save are written there, and the import-query dialog opens there. It is created automatically; you do not need to make it yourself.

3. PC-ACE data

PC-ACE tables exported to Excel are ordinary .xlsx files, so they are valid input: point the input directory at the folder of exported PC-ACE tables and the GUI builds the database from them, applying the standard PC-ACE column renames as it goes.

Because of this, you can open the DB SQL GUI directly from the PC-ACE Tables Analyzer or the PC-ACE Data Validation GUI (Open DB SQL GUI in their dropdown). Those GUIs simply open this one; this GUI then builds and validates its own database. There is no separate export step to run first.

See also the TIPS "PC-ACE - Export ACCESS tables to Excel" (how to get the .xlsx files out of PC-ACE) and "PC-ACE table relations" (how the tables join).

4. Running queries

Step	What to do
Pick the database	Set the input directory (the database builds itself), or select an existing .sqlite file.
Write a query	Type SQL in the query area, or start from a template (see the "SQL template queries" TIPS).
Run it	Press RUN. Results are written to your output directory.
Save / re-import	Saved queries go to the "SQL queries" subdirectory of the input directory.

If a query fails, the GUI reports the SQLite error. The most frequent cause is a table or column name that does not exist in YOUR database — template queries in particular must be edited so their table and column names match your data. Use the table/field dropdowns to check the real names.

5. Troubleshooting

Symptom	Cause and fix
All widgets greyed out; nothing happens	No SQLite database is available. The input directory has no .xlsx/.csv files (a text corpus was probably selected), or the input/output directory is unset. Select a folder of data files, or select an existing .sqlite.
"The query did not execute properly"	A table/column name in the query does not match your database. Edit the query to use your real names — especially with template queries.
Database seems out of date	Delete the .sqlite file in the input directory and re-select the folder to force a clean rebuild.
Build takes a long time	Normal for a large set of tables. The title bar shows "Loading ..." while it works; it is reused (not rebuilt) next time.